

2. 4-bit Multiplier :-

> Multiplication functionality: $\rightarrow \text{Product} = A \times B$
 $\downarrow \quad \quad \quad \downarrow \quad \quad \downarrow$
 8 bit 9 bit 4 bit 4 bit

→ Ask for pro flag for multiply by 0.

→ Maximum product conditions:

$$15 \times 15 = 225$$

$$1111 \times 1111 = 11100001$$

> Directed tests:-

→ multiply by 1.

→ multiply by 0.

→ Multiply by some number

> Turner cases:-

$$\rightarrow A=15, B=0$$
$$\rightarrow A=0, B=15$$
$$\rightarrow A=15, \theta=1$$

→ $A = 15$, $B = 15$

$$\rightarrow A = 1000, B = 1000$$
$$\rightarrow A = 0100, B = 0100$$

$\rightarrow A = 0010$, $B = 0010$

> Functional coverage :-

$\rightarrow \text{bin } A = \{0:15\}$, $\text{bin } B = \{0:15\}$

→ cross coverage of A, B

→ Product coverage {0:255}

> Reference model: $P_{ref} = a * b$

$$P_{ref} = P_{det}$$